AgriPollinate

TEXAS A&M UNIVERSITY

Jason Agnew | Paavan Bagla | Josiah Faircloth | Samuel Ramos

Sponsor: SICK | Faculty: Stavros Kalafatis

TEXAS A&M UNIVERSITY Department of Electrical & Computer Engineering

SICK

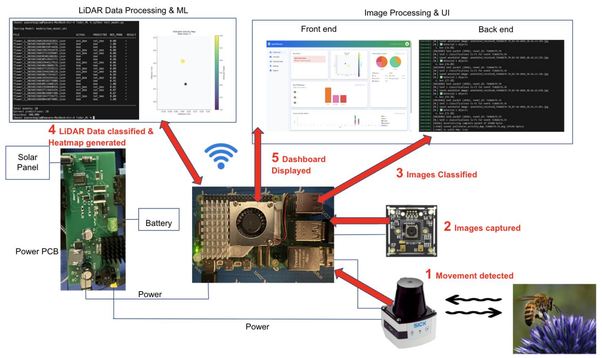
Problem Definition

Farmers lack an efficient and scalable way to monitor pollination, relying on manual and inconsistent methods.

AgriPollinate uses LiDAR and image based detection to automatically track and classify pollinator activity, displaying results through heatmaps on a dashboard.

The portable, solar powered system enables real time field deployment and improved decision making.

Methodology



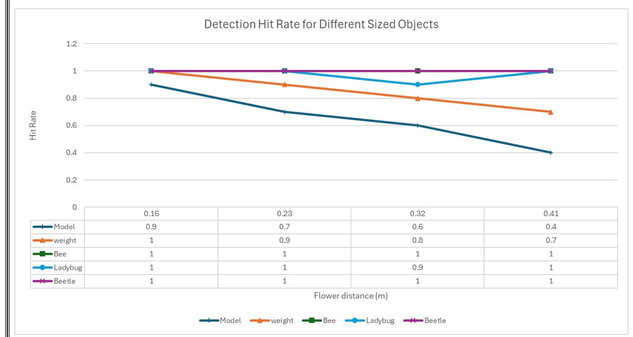
- **Setup:** Flowers are configured with GPS coordinates in the system and dual TCP connections are established while the devices are powered by a PCB.
- **Detection:** The LiDAR continuously scans flowers for movement activity.
- **Event Trigger:** When movement is detected across 6 consecutive scans, an event is triggered and the camera captures an image of the flower.
- **Classification**: LiDAR data is sent to the ML model; camera image is sent for visual classification. Both are processed simultaneously.
- **Dashboard:** Displays LiDAR heatmap, insect species counts, top 3 pollinators, insect visits per each hour of the day, and triggers pest alerts.

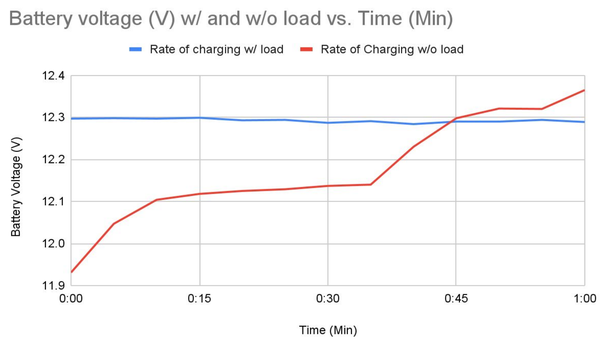
Engineering Analysis

Throughout the course, our project changed in many ways, our scope, data collection methods, and power had to be refined to match our goals. System validation was performed across hardware and software subsystems. Detection accuracy was verified within a set detection range under field conditions. The ML classification models were trained and evaluated on labeled pollinator image datasets. Finally, the PCB power charging rate was validated with and without system load.

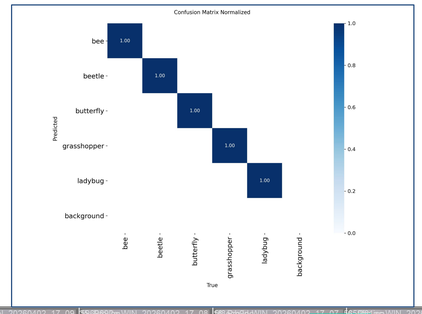
Performance Metrics for Rasp. Pi

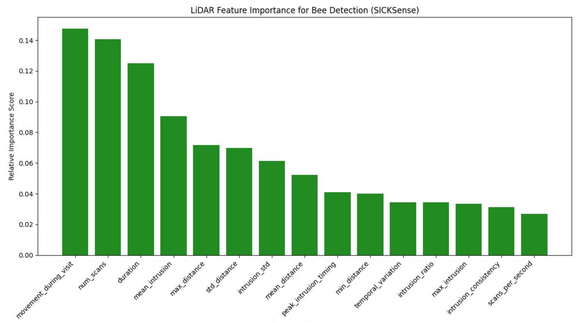
Requirement	Spec	Result	Status
Transmission Latency	<500 ms	avg. 197 ms (25 trials)	met
End to End Latency	< 5 s	avg. 3.9 (25 trials)	met
Detection Performance	70-80% accuracy	See Graph	met
Auto TCP Reconnect	Functional	Verified in full system testing	met
Memory Efficiency	< 85% = 10 GB	avg. 7.4 MB (12 cycles)	met
Fault Tolerance	Functional	Verified in full system testing	met





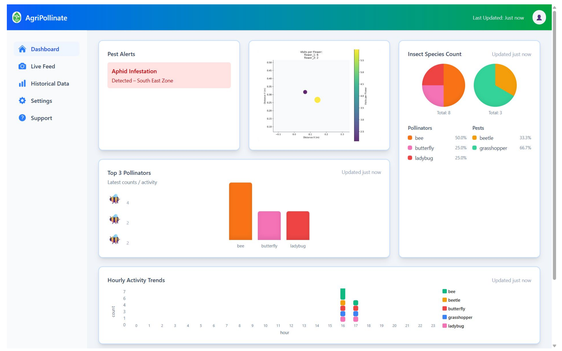
Outcomes





Per-group test results

Group	Description	Pass/Total	Acc
T0	Leaf Falling (<1sec)	**3/3**	100%
T1	Bee vibrating 1-4s	**8/8**	100%
T2	Bee vibrating 4-8s	**5/6**	83%
T3	Bee still 2-5s	**3/4**	75%
T4	Short Not-bee still 2-6s	**7/8**	87.5%
T5	Not-bee still 8-15s	**9/9**	100%
T6	Long Not-bee still 15-30s	**3/3**	100%



Impact

Our project serves as a proof of concept model for a pollination monitoring system which can provide farmers with real-time pollination data to make faster, more informed decisions and reduce reliance on manual monitoring, cutting labor costs and human error. Heat-map visualization enables spatial analysis of pollinator hotspots across fields which can inform hive placement adjustments and solar-powered portability makes deployment viable for small and large-scale farms alike.

References

- https://www.sciencedirect.com/science/article/pii/S2214574525000379

Acknowledgments

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